Five 1-in-1000-year floods in one month: what are we learning from it?

Zhi (Allen) Li, PhD

li1995@ou.edu

Postdoctoral researcher
Hydrometeorology and Remote Sensing Laboratory
University of Oklahoma

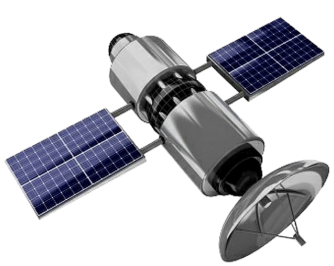
Date: 08/31/2022



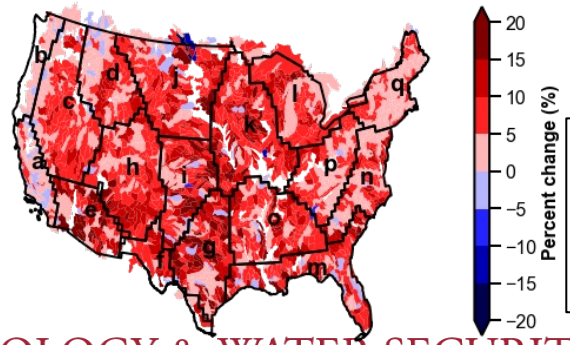
HYDROLOGY & WATER SECURITY PROGRAM
The UNIVERSITY of OKLAHOMA

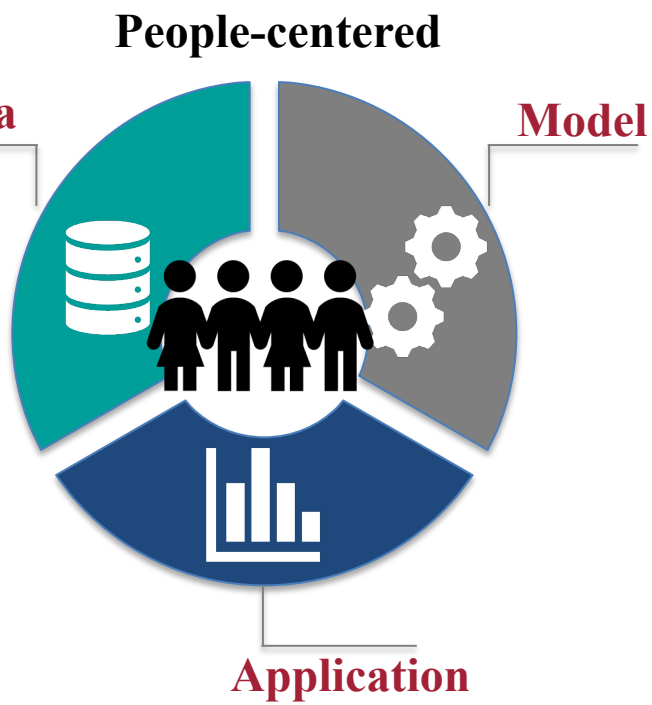
PhD dissertation:
Advancing flood prediction and projection in a warming climate

1. **Remote sensing** monitoring water cycle (Li et al., 2021; Gao et al., 2021)
2. A **120-year flood database** to analyze flood risks (Li et al., 2021)

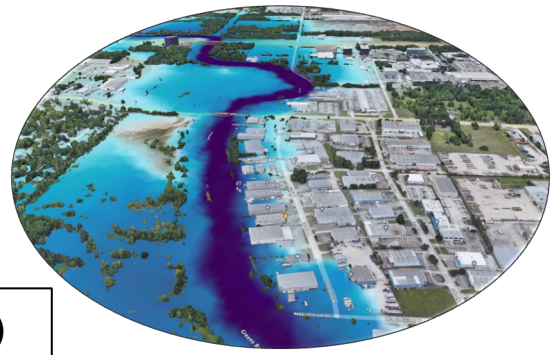








1. Developed a **hydrologic-hydraulic coupled model** for better flood prediction (Li et al., 2021,2022)
2. Developed a vector-based hydrologic model for **large-scale** streamflow prediction (fast and high resolution) (Li et al., 2022)



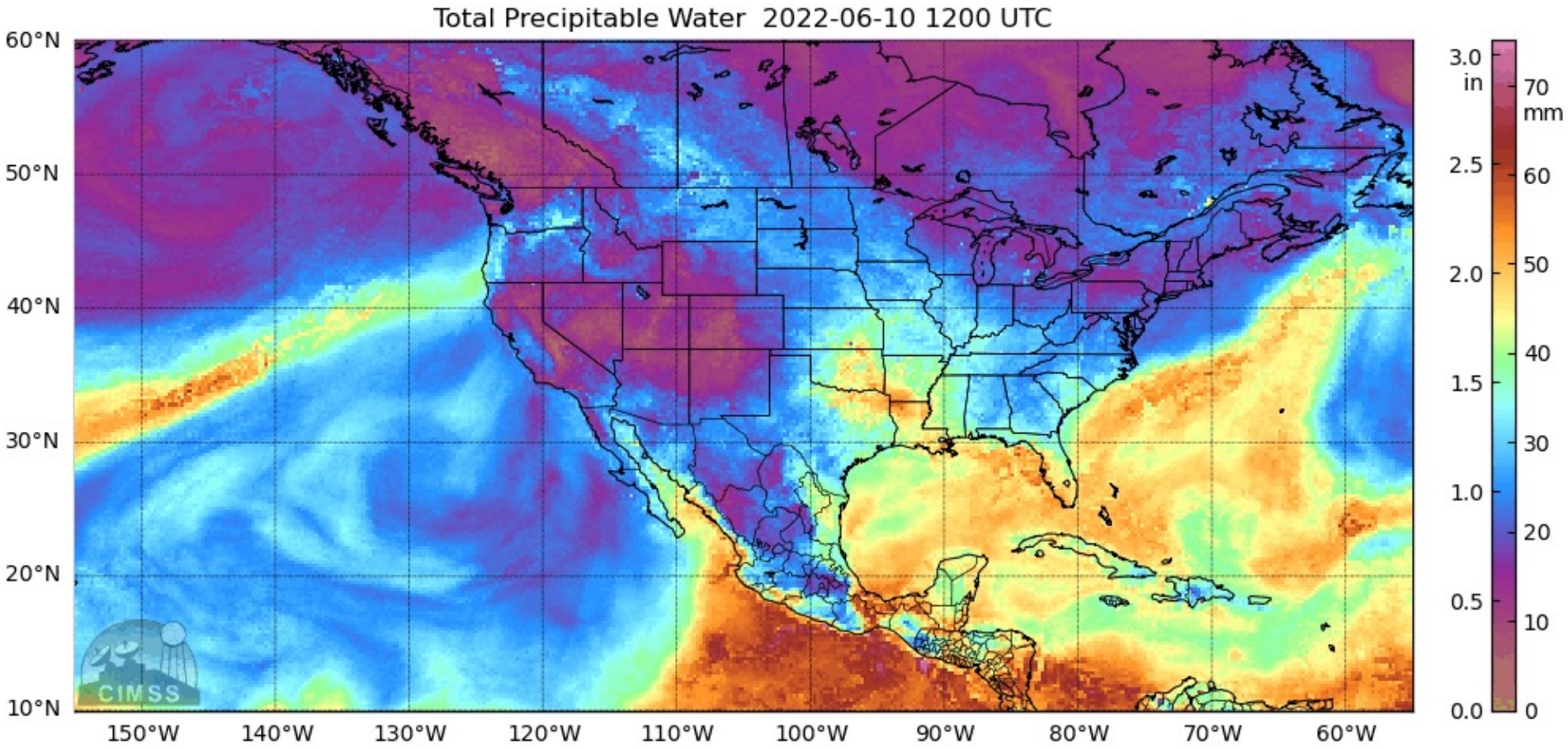
1. Investigated **future flash floods** (Li et al., 2022)
2. **Spatiotemporal characteristics** of future floods (Li et al., 2022)

Five historical floods in five weeks!



Source: AccuWeather

Yellowstone flooding – Atmospheric river



Flash flood events associated with high PW values (Maddox et al., BAMS, 1979)

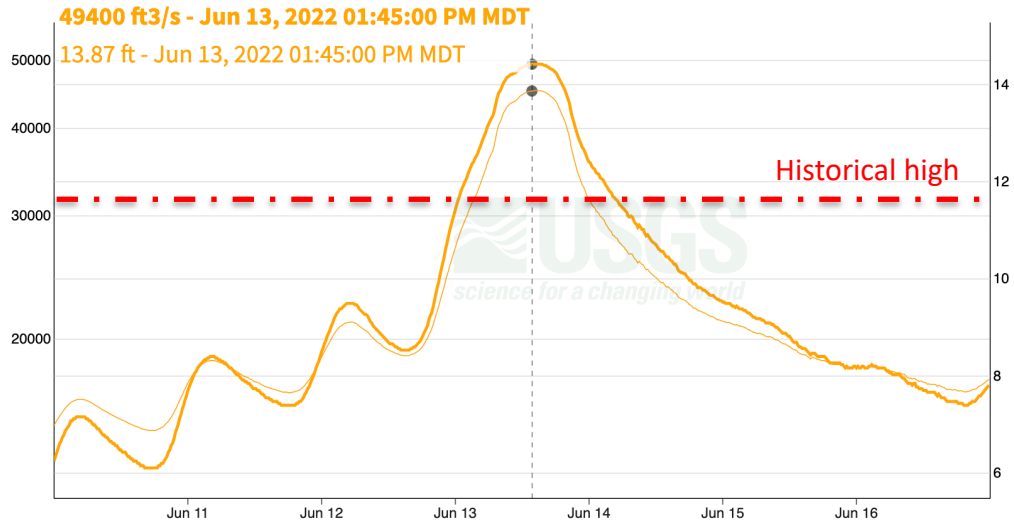
Yellowstone flooding – rain-on-snow event



Snow cover in June 2021



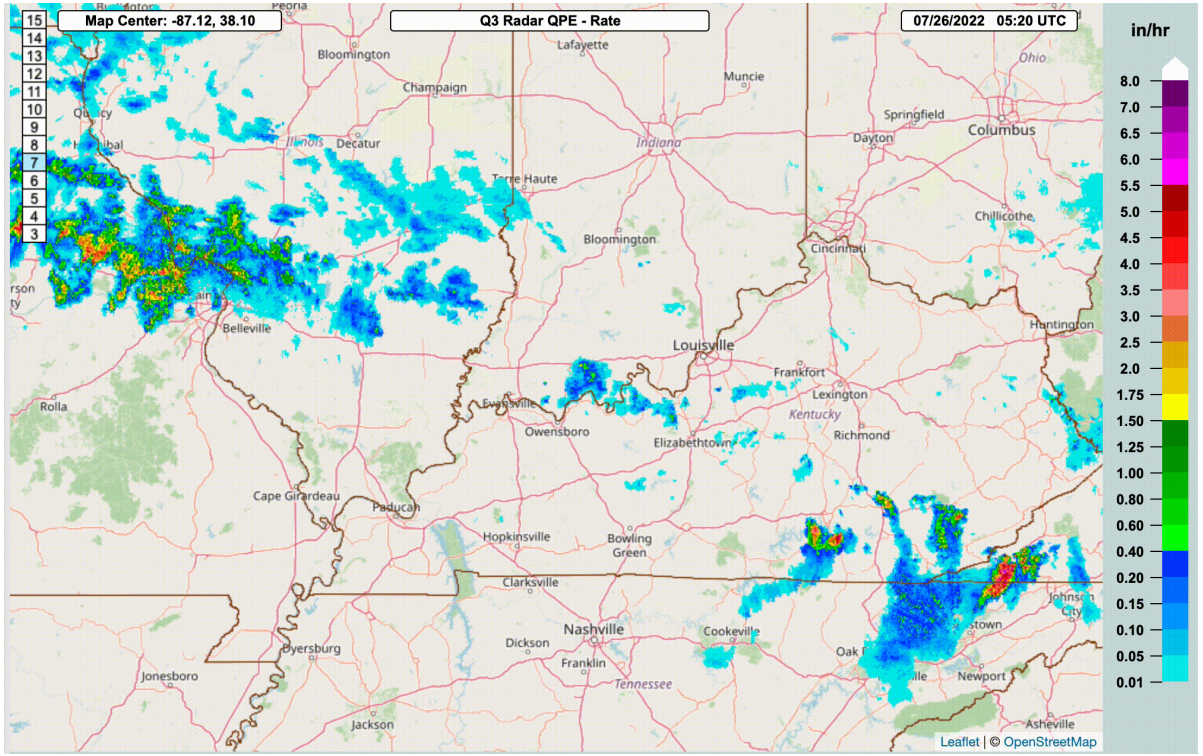
Snow cover in June 2022



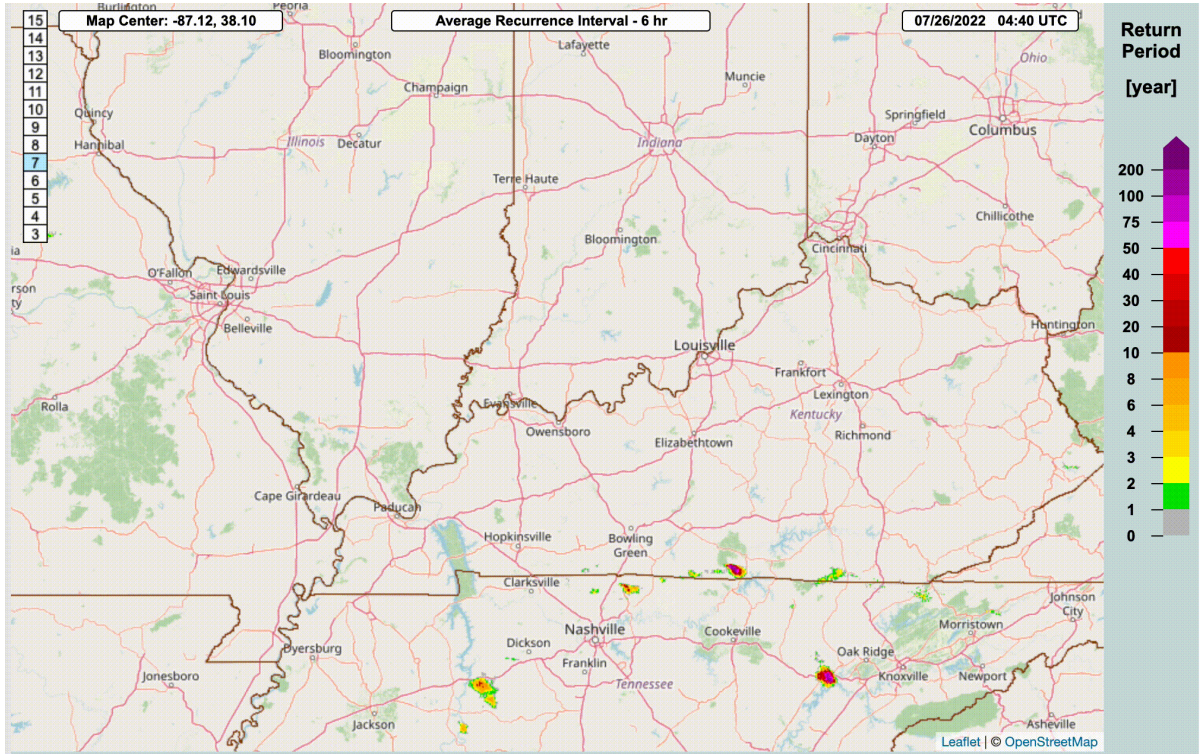
Source: USGS

Source: https://earthobservatory.nasa.gov/images/150010/catastrophic-flooding-in-yellowstone

St. Louis/Kentucky flooding – training thunderstorms



Source: MRMS



Source: CREST unit flow

12’’ in 12 hours!

St. Louis/Kentucky flooding – steep terrain

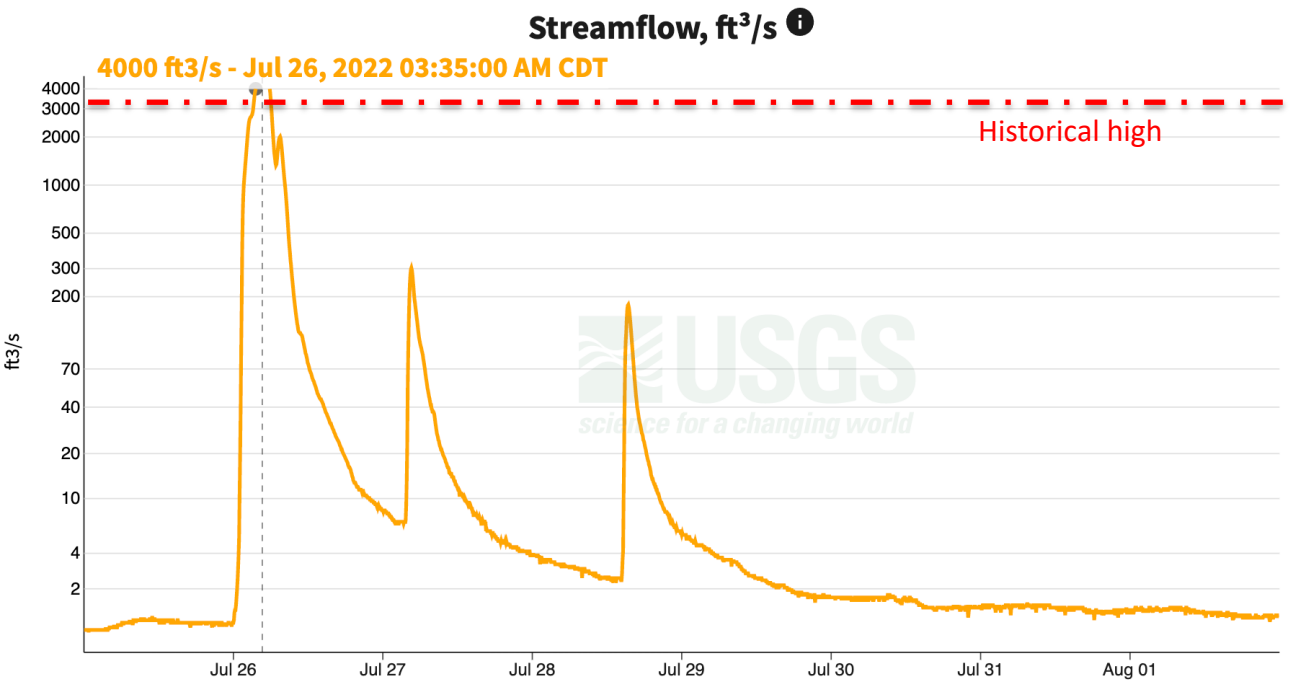


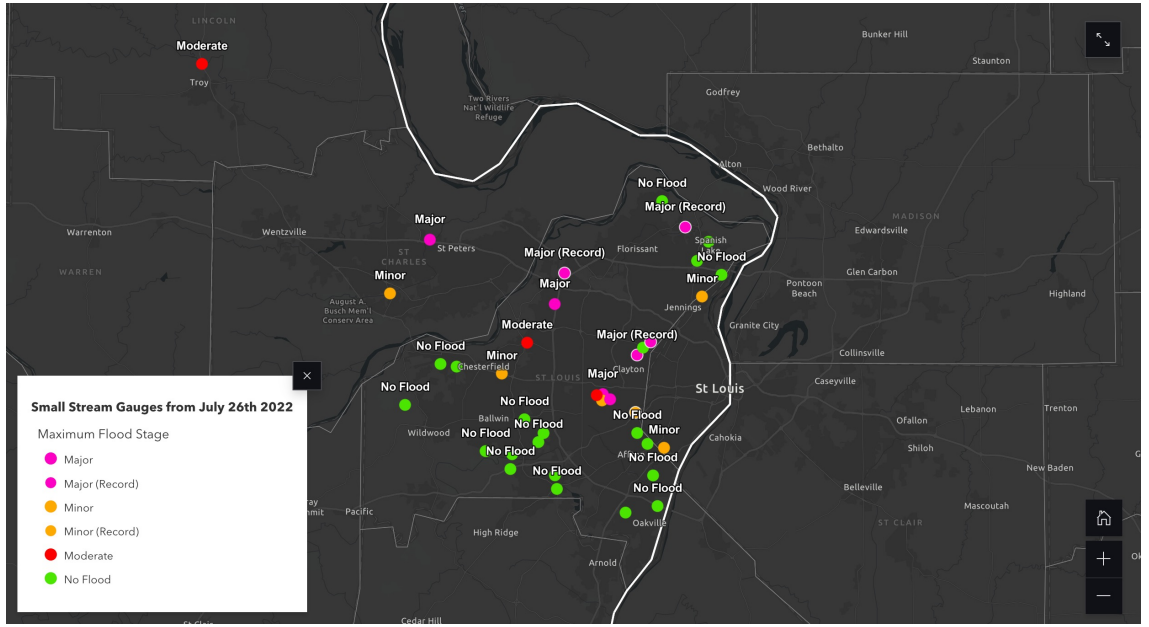
Jackson, Kentucky



Hazard, Kentucky

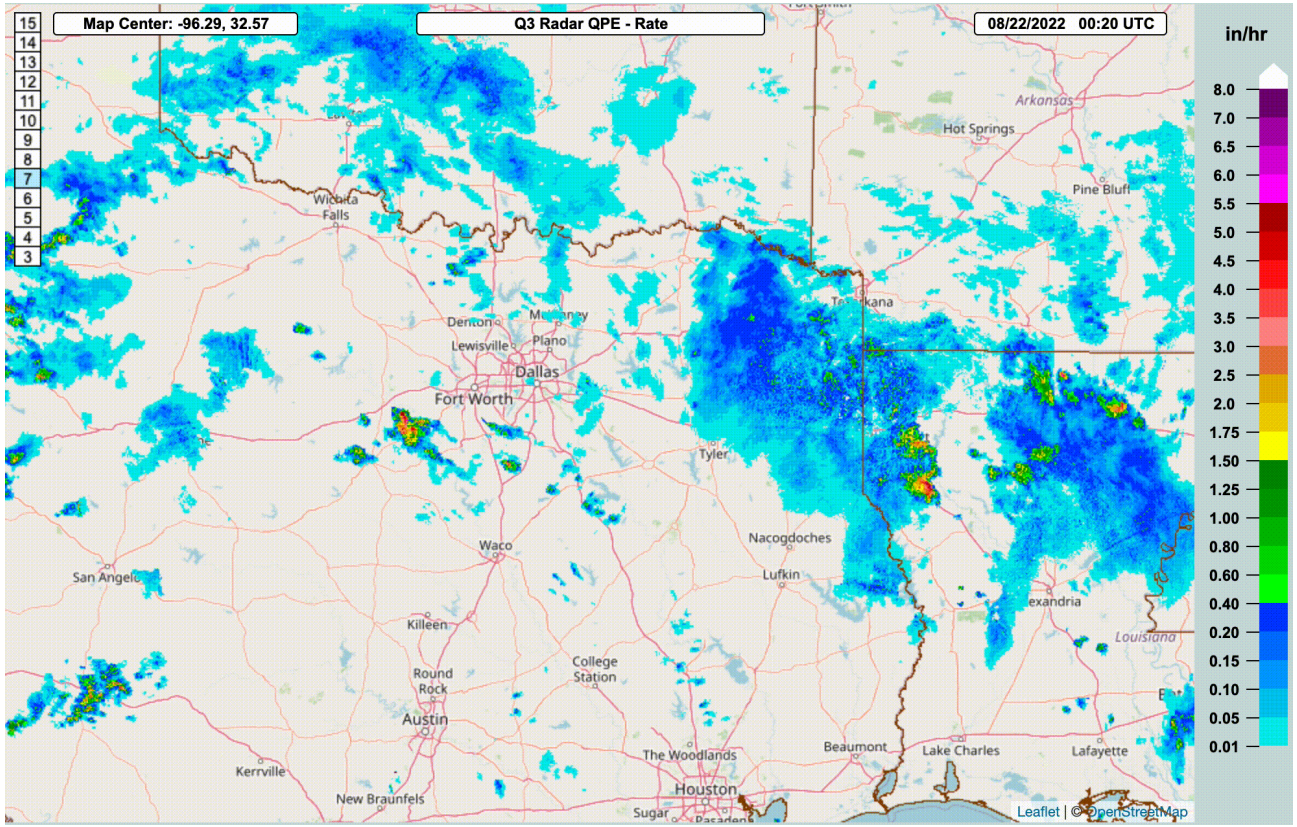
St. Louis/Kentucky flooding – record-breaking streamflow





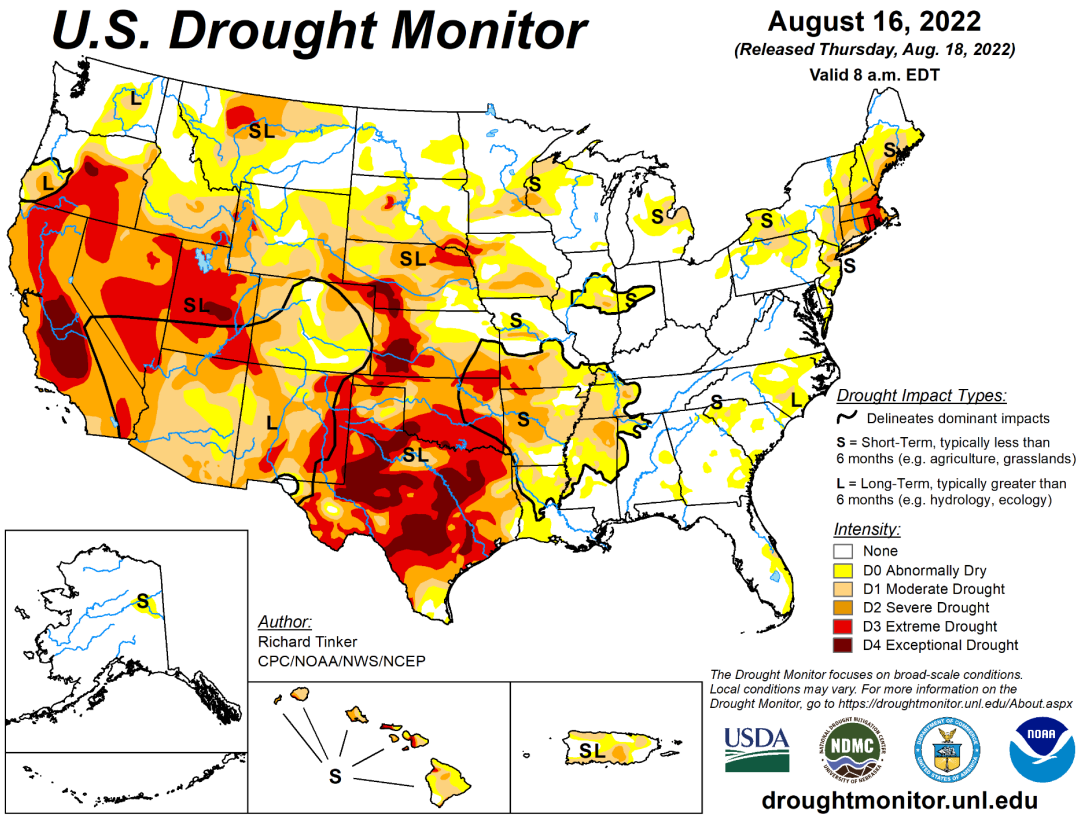
Source: https://storymaps.arcgis.com/stories/9d10335079444c159966e0a28c90c4df

Dallas flooding – training thunderstorms



Over 3 inches in one hour
10 inches in 12 hours
15 inches in 24 hours

Dallas flooding – antecedent drought exacerbates flood condition





Credit: University of Reading/Dr. Rob Thompson

drought-pluvial seesaw! (He et al., 2020)

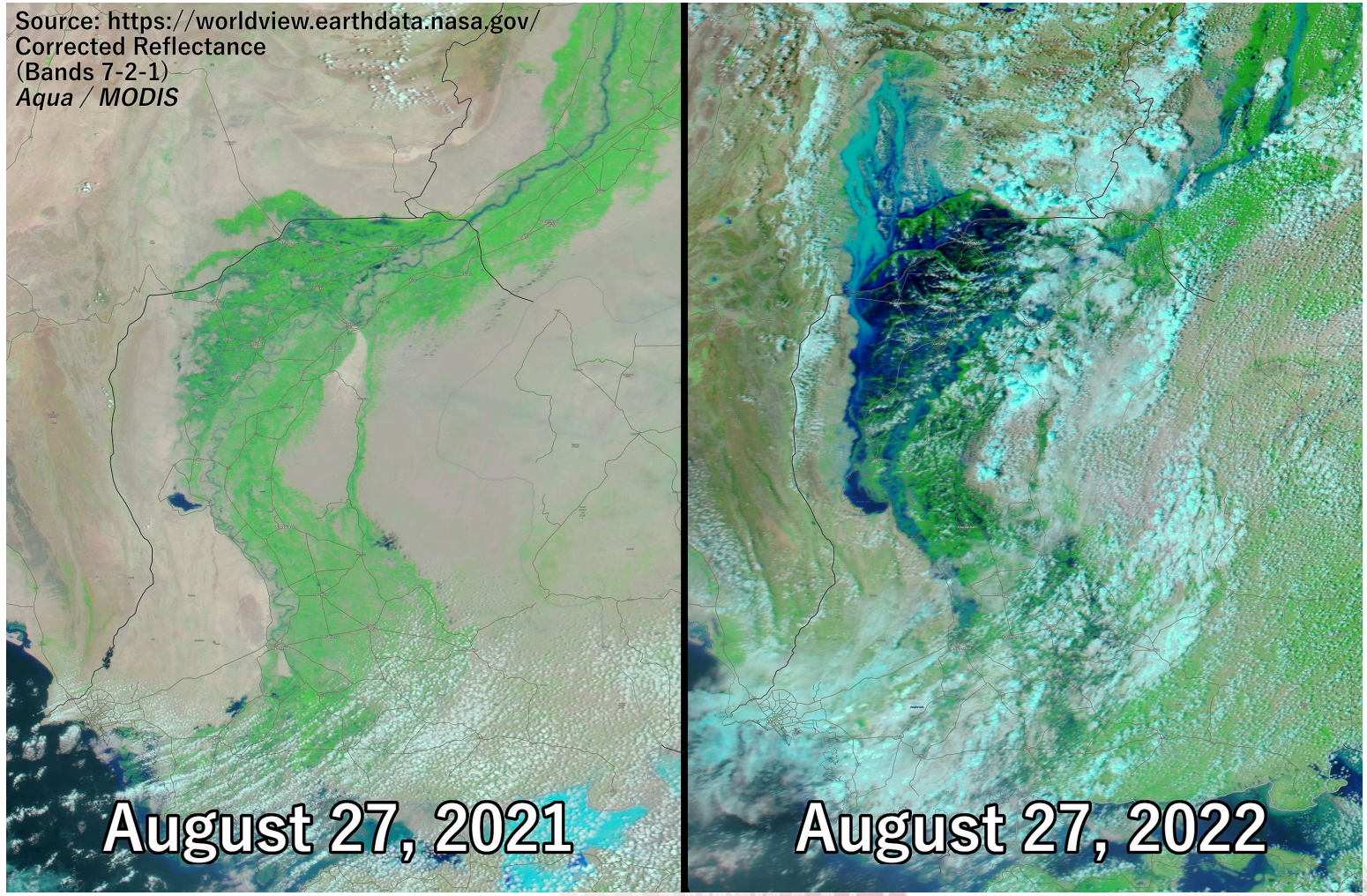
Pakistan flooding

Unusual Monsoon rainfall + melting glaciers

Thousands of people killed

30 million people displaced

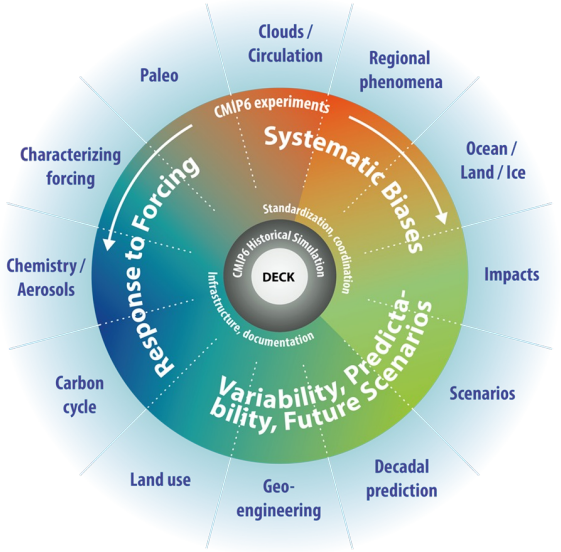
700k homes destroyed



What does it tell us?

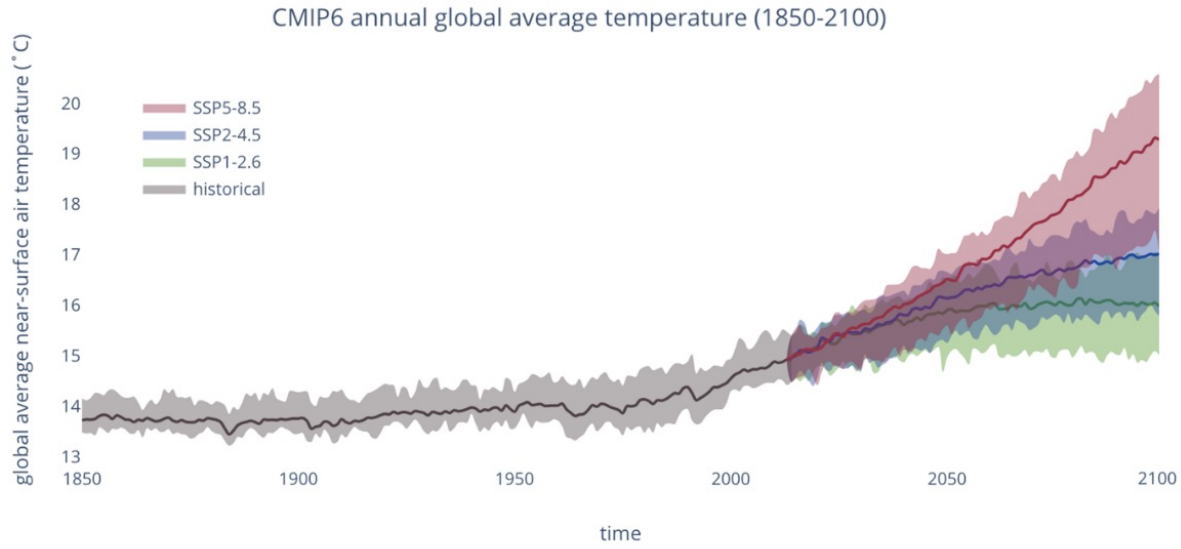
- Atmospheric condition + land surface condition jointly cause these devastating events
- What is the role of climate change?

Challenges in flash flood projection



1. Desire a high resolution climate projection data set
2. Uncertainties in climate internal variabilities
3. Lack of continuous streamflow simulation in modern climate models

FEW studies have quantified changes in flash floods



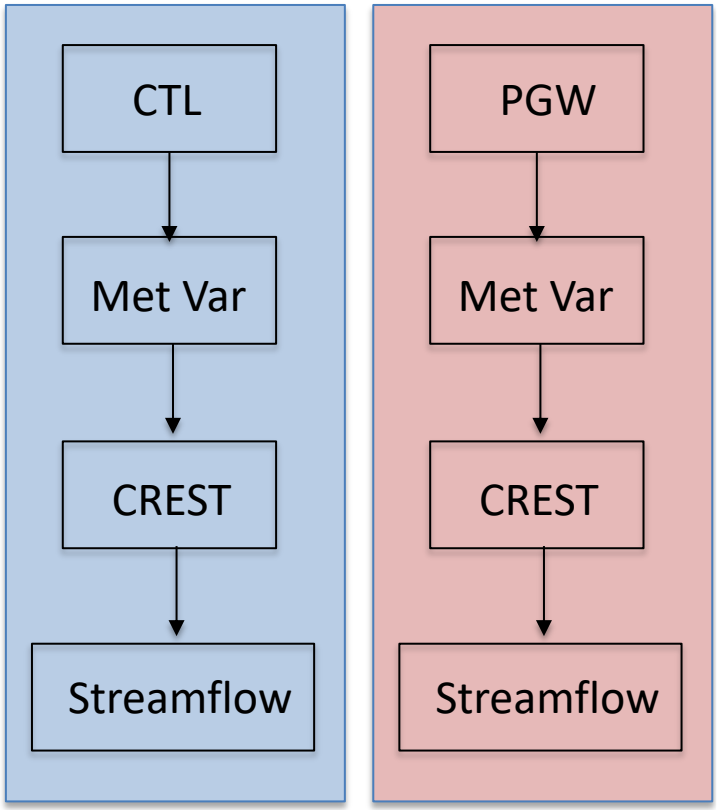
Credit: Copernicus Climate Change Service, ECMWF.

Climate projection - dynamic downscaling

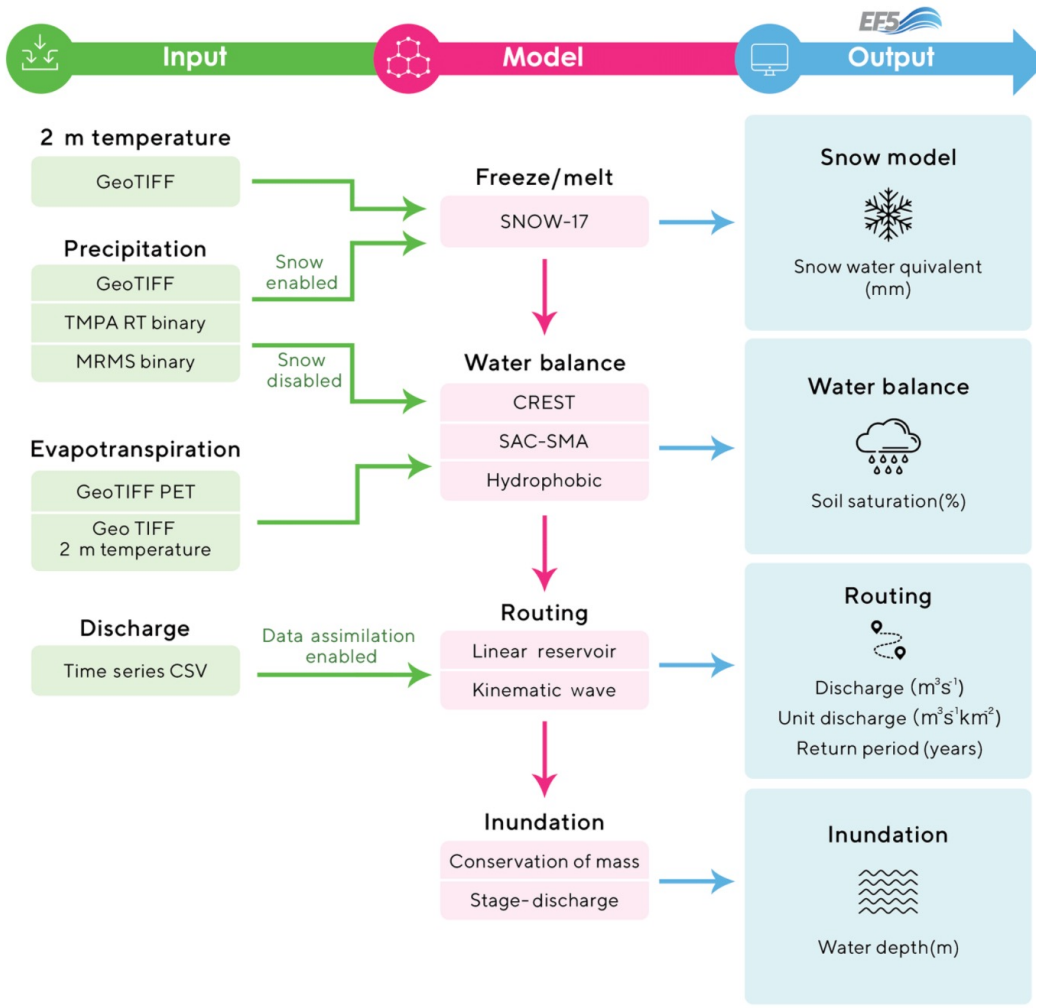
High-res Climate simulation over the CONUS (at **hourly** and **4-km** resolution, convection-permitting) (Liu et al., 2017):

- Retrospective simulation forced by 6-hourly ERA-Interim reanalysis data (**CTL**), 2000 - 2013
- Pseudo-global warming (**PGW**): PGW = ERA-Interim + $\Delta CMIP5_{RCP8.5}$
  - $\Delta CMIP5_{RCP8.5} = CMIP_{2071\text{-}2100} - CMIP_{1976\text{-}2005}$

- PGW perturbs the *horizontal winds*, *geopotential*, *temperature*, *humidity*, *SST*, *soil temperature*
- Spectral nudging is applied to reproduce synoptic weather events



Hydrologic simulation – 1km continental streamflow



Flamig et al., 2020

Simulation period: 2001 – 2013

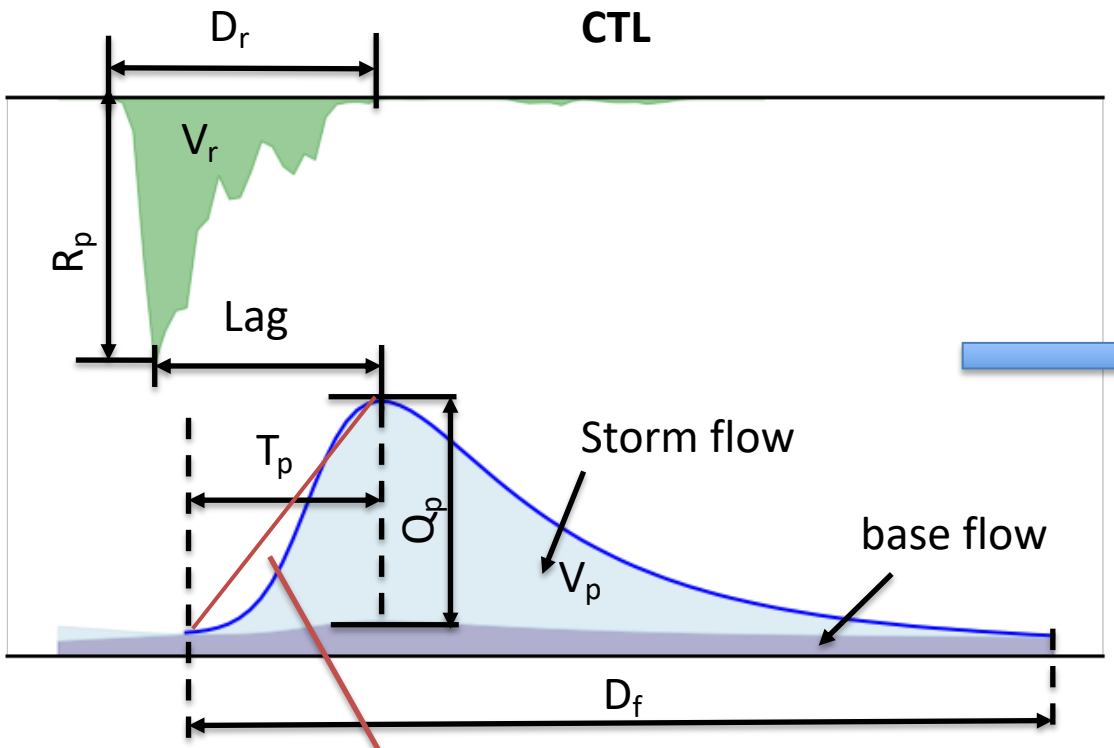
Model: Snow 17 + CREST

Resolution: 1-km and hourly

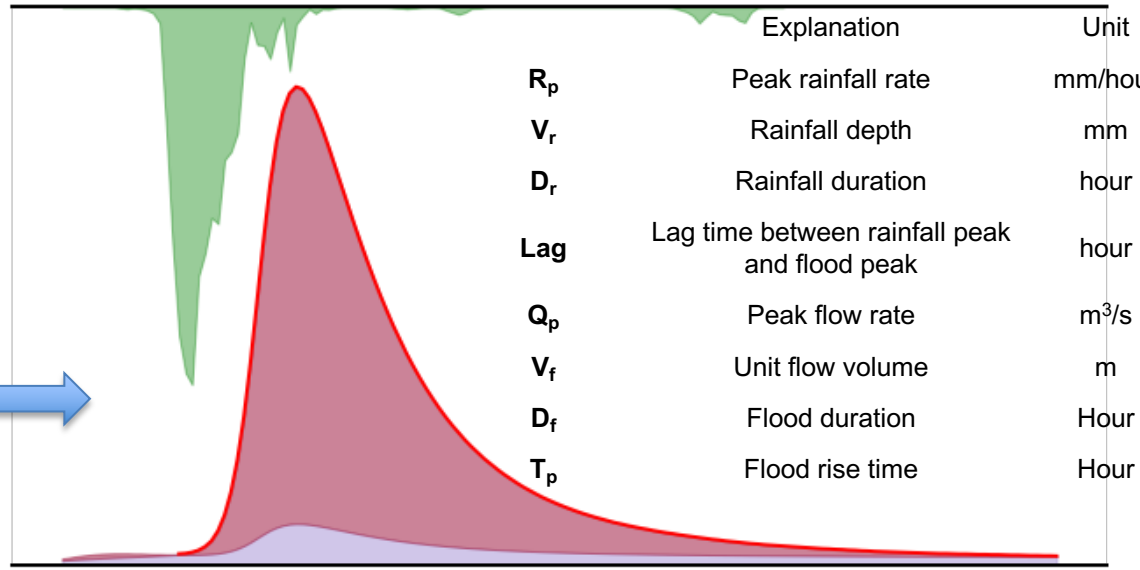
Inputs: precipitation, temperature from CTL and PGW

Outputs: Streamflow and soil moisture

Quantify flashiness



PGW



	Explanation	Unit
R_p	Peak rainfall rate	mm/hour
V_r	Rainfall depth	mm
D_r	Rainfall duration	hour
Lag	Lag time between rainfall peak and flood peak	hour
Q_p	Peak flow rate	m^3/s
V_f	Unit flow volume	m
D_f	Flood duration	Hour
T_p	Flood rise time	Hour

$$Flashiness_{s,i,j} = \frac{Qp_{s,i,j} - Qb_{s,i,j}}{A_i T_{s,i,j}}$$

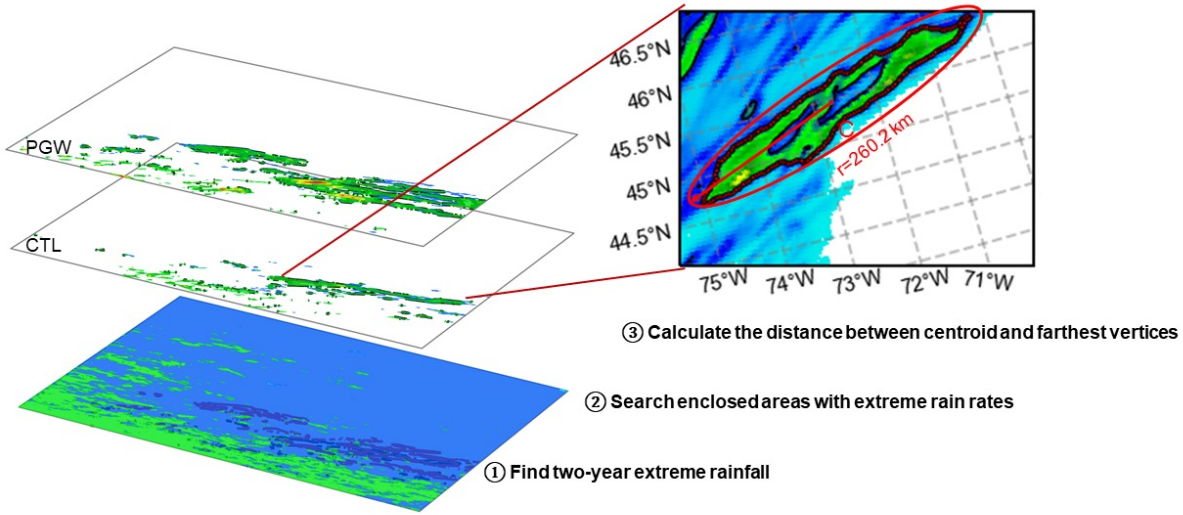
↑ High magnitude

↑ Short time

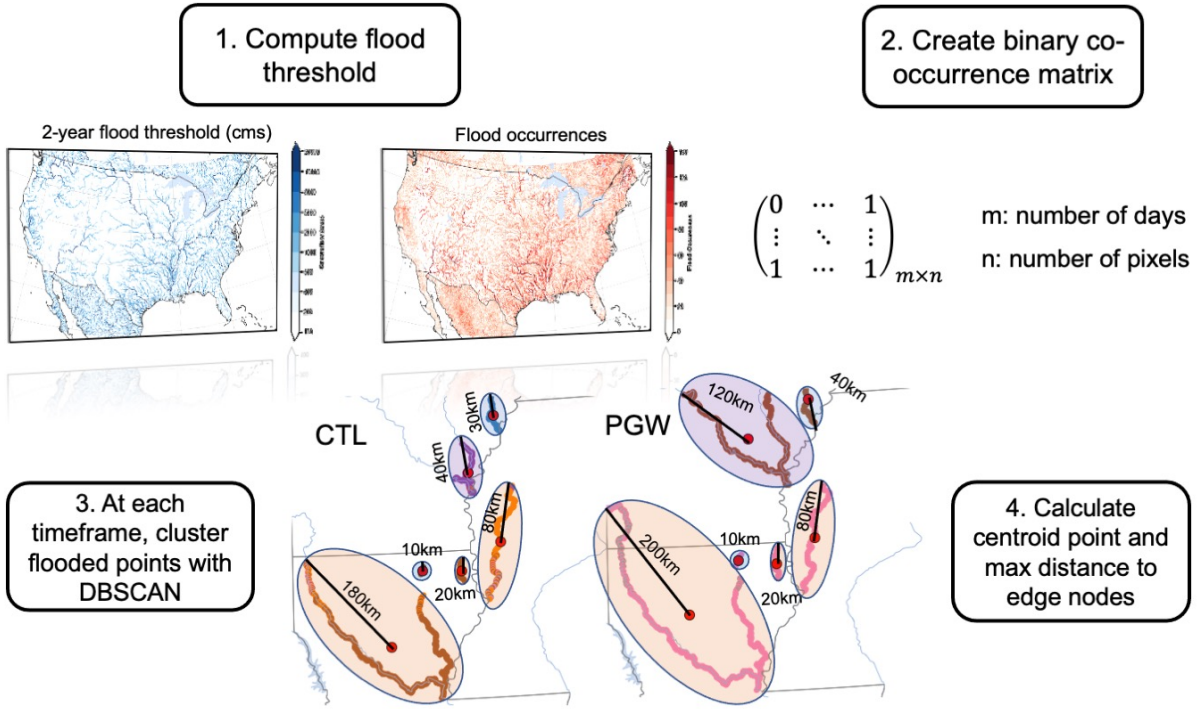
Li et al., 2022, Commun. Earth Environ.

Quantify spatial scales of rainfall and floods

Rainfall spatial scale



Flood spatial scale

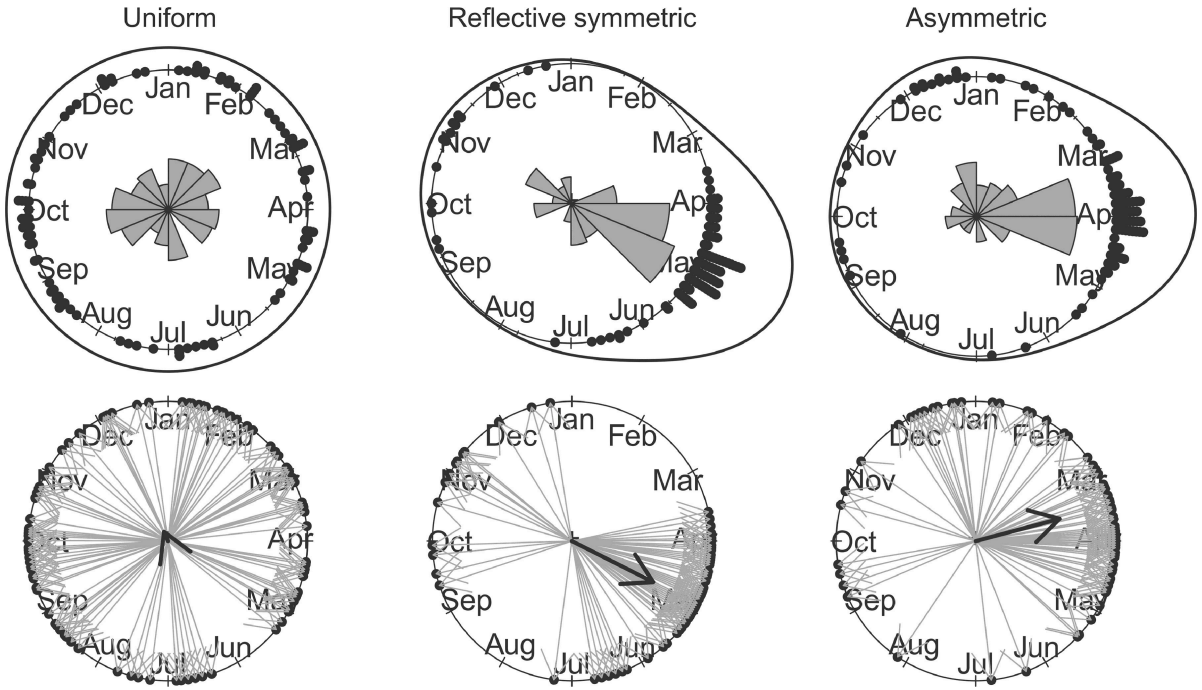


Li et al., 2022, Earths Future

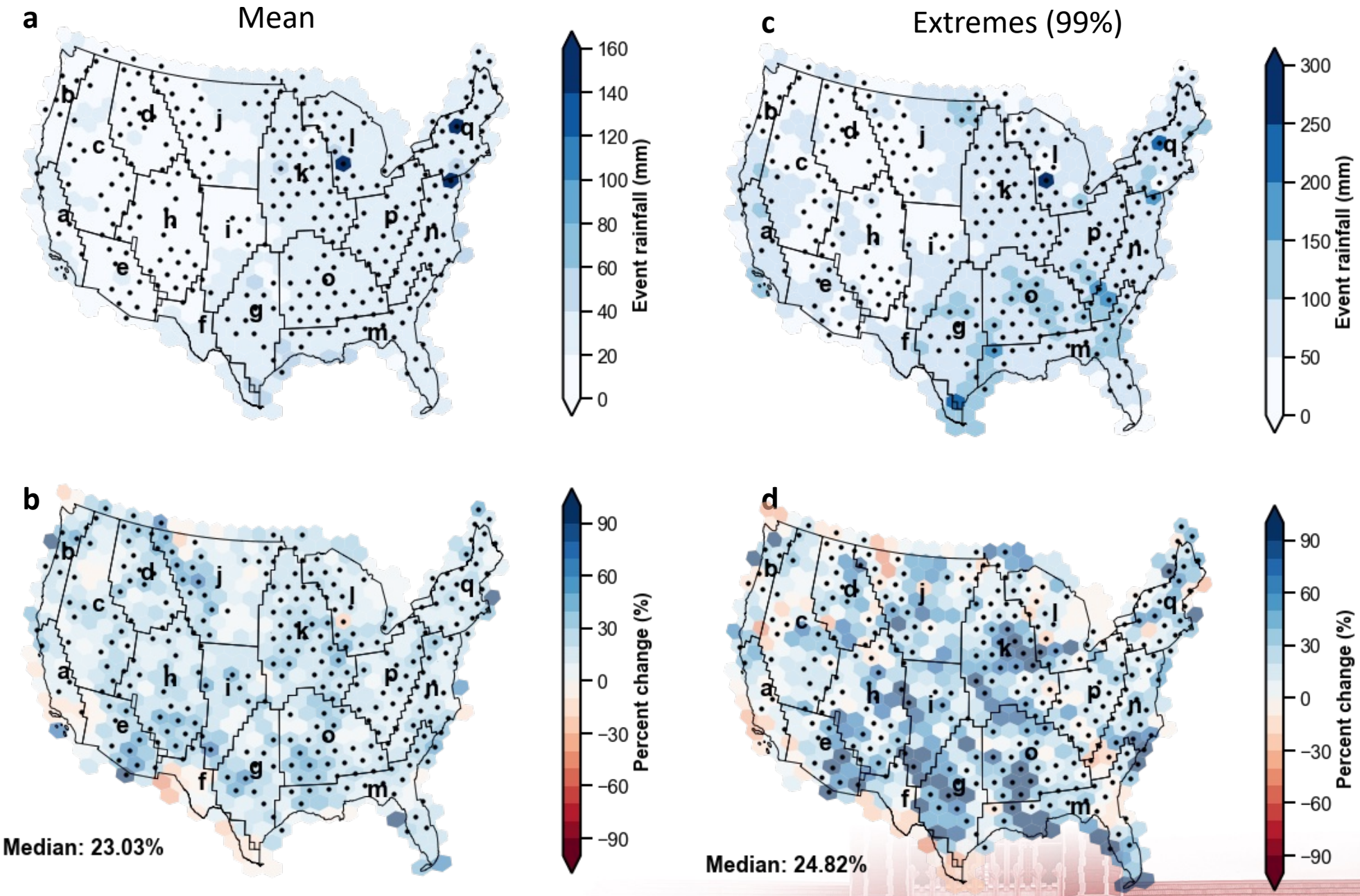
Quantify seasonality

Seasonality

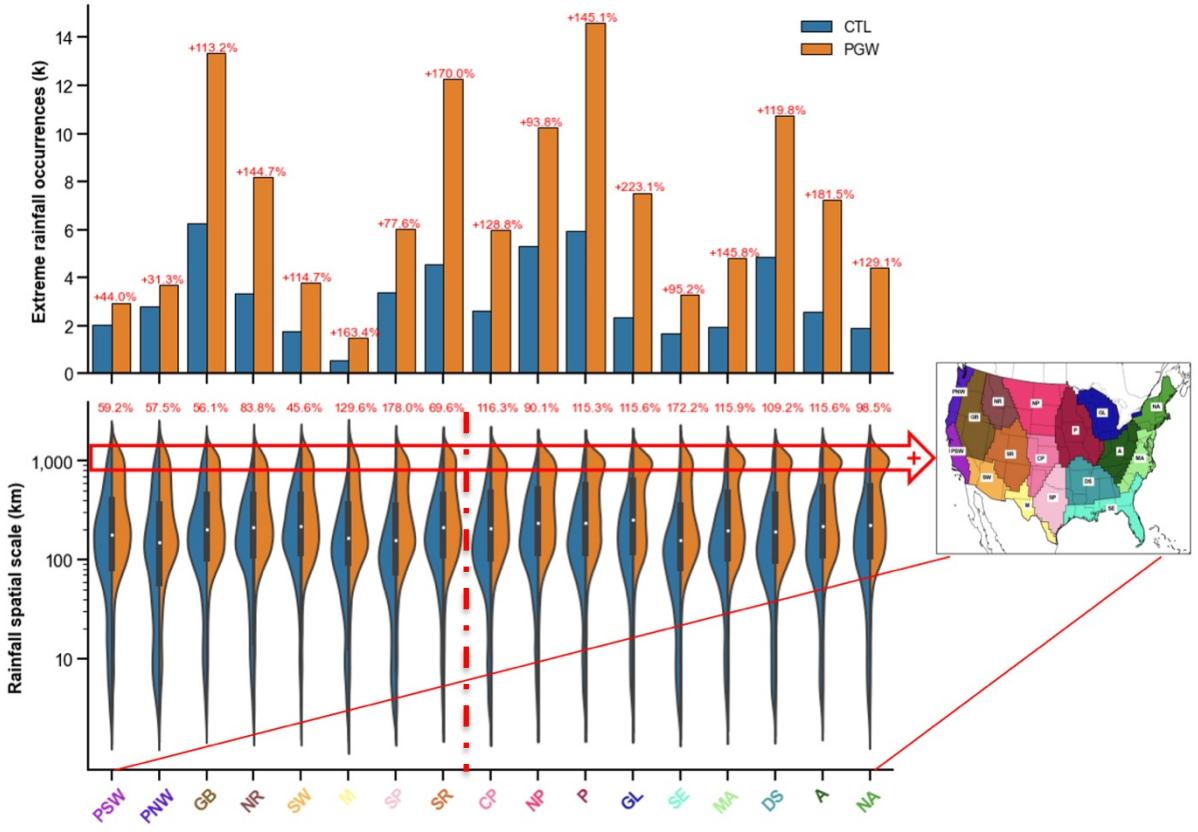
Strength of seasonality

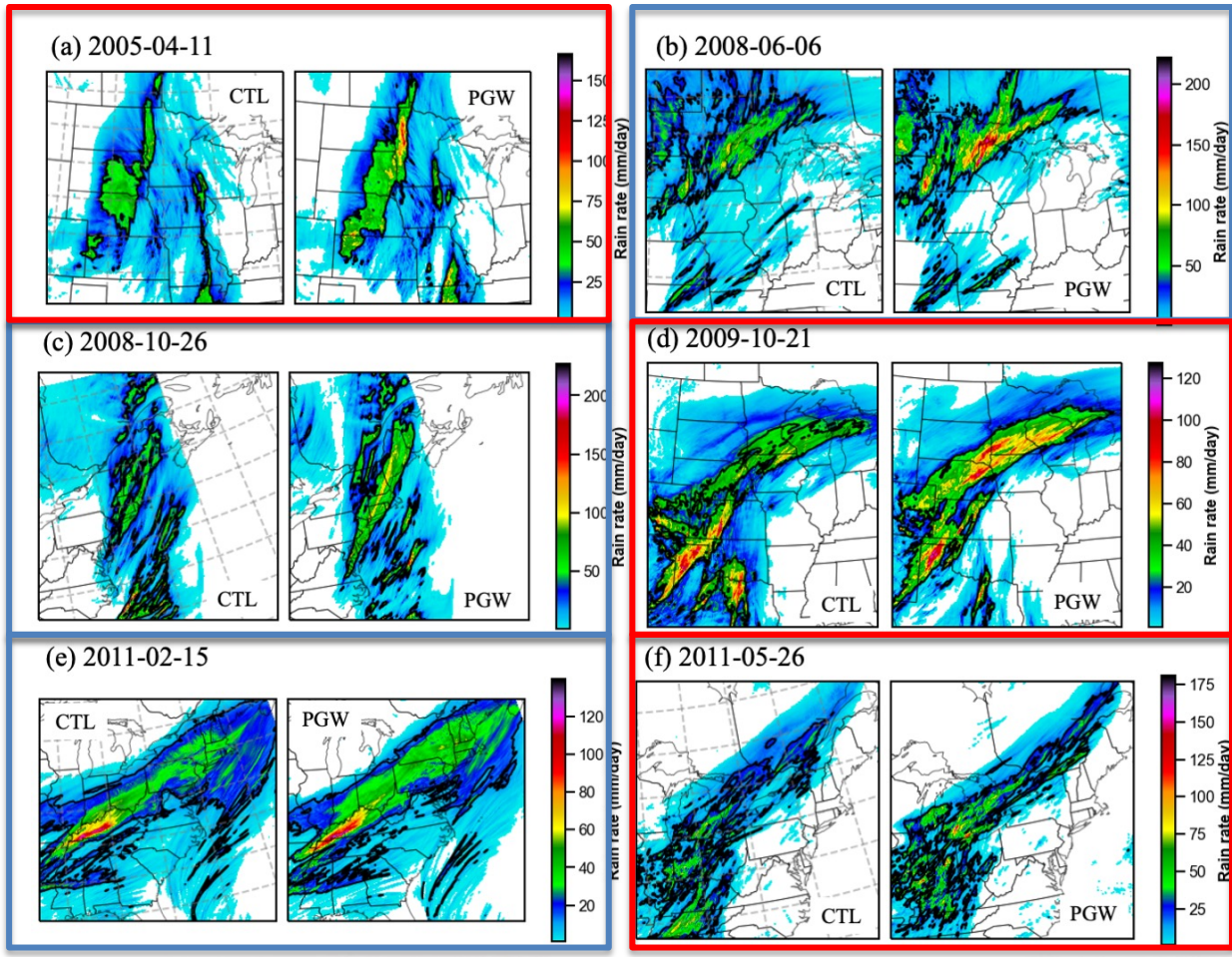


Credit: Villarini (2016)


a
Mean
160
140
120
100
80
60
40
20
0
Event rainfall (mm)
c
Extremes (99%)
300
250
200
150
100
50
0
Event rainfall (mm)
b
90
60
30
0
−30
−60
−90
Percent change (%)
Median: 23.03%
d
90
60
30
0
−30
−60
−90
Percent change (%)
Median: 24.82%

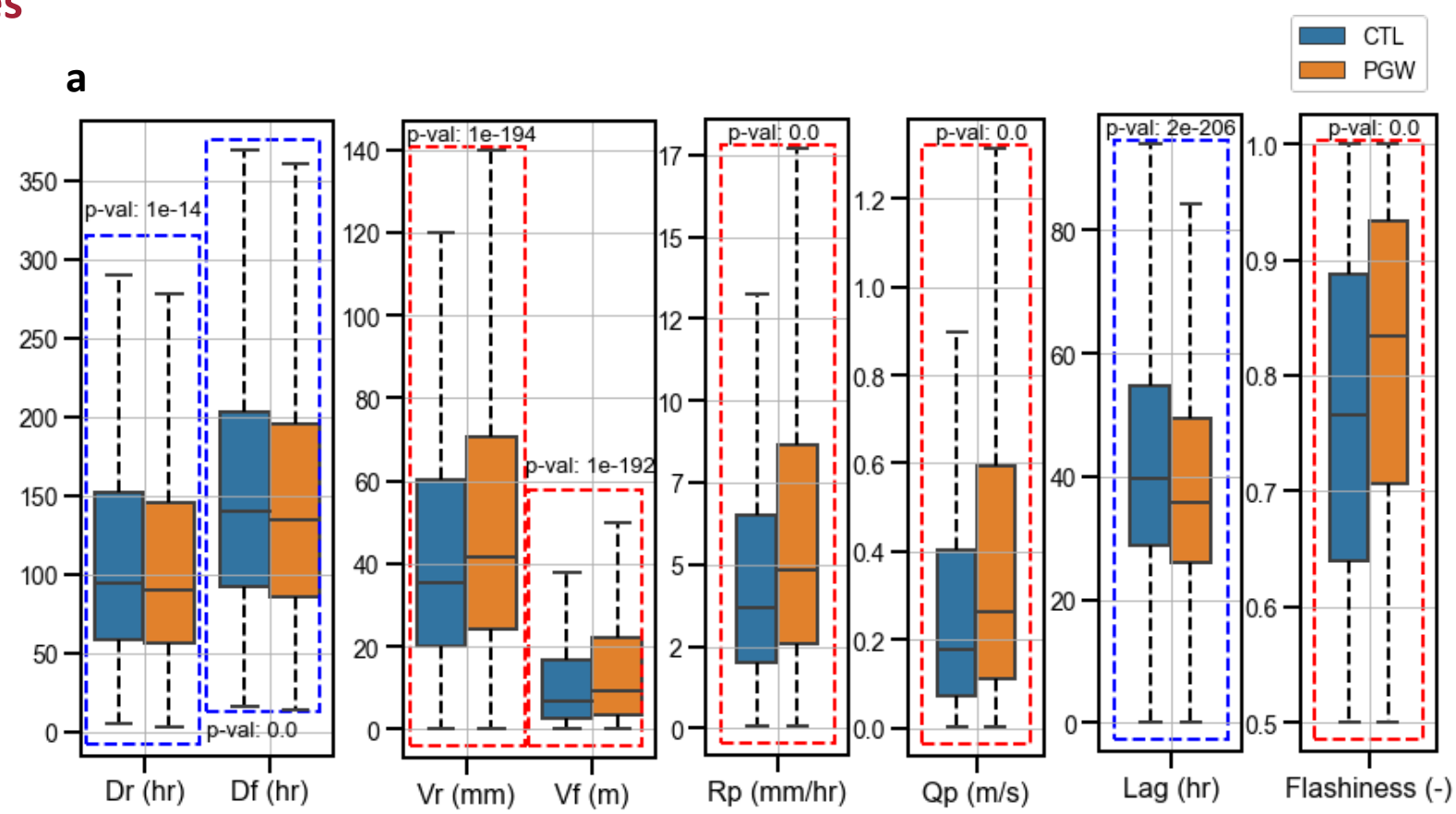
spatial scales of rainfall



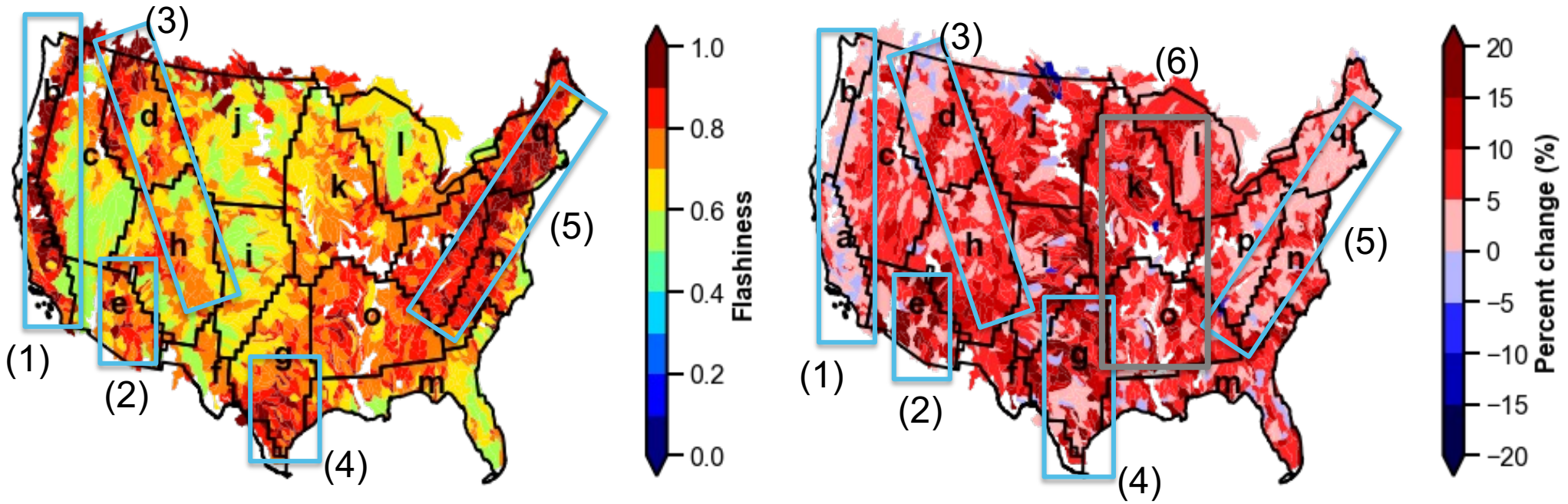


Changes in flow signatures

D_r: Rainfall duration
D_f: flood duration
V_r: rainfall volume
V_f: flood volume
R_p: Rainfall rates
Q_p: Streamflow
Lag: Rainfall-to-flood time lag



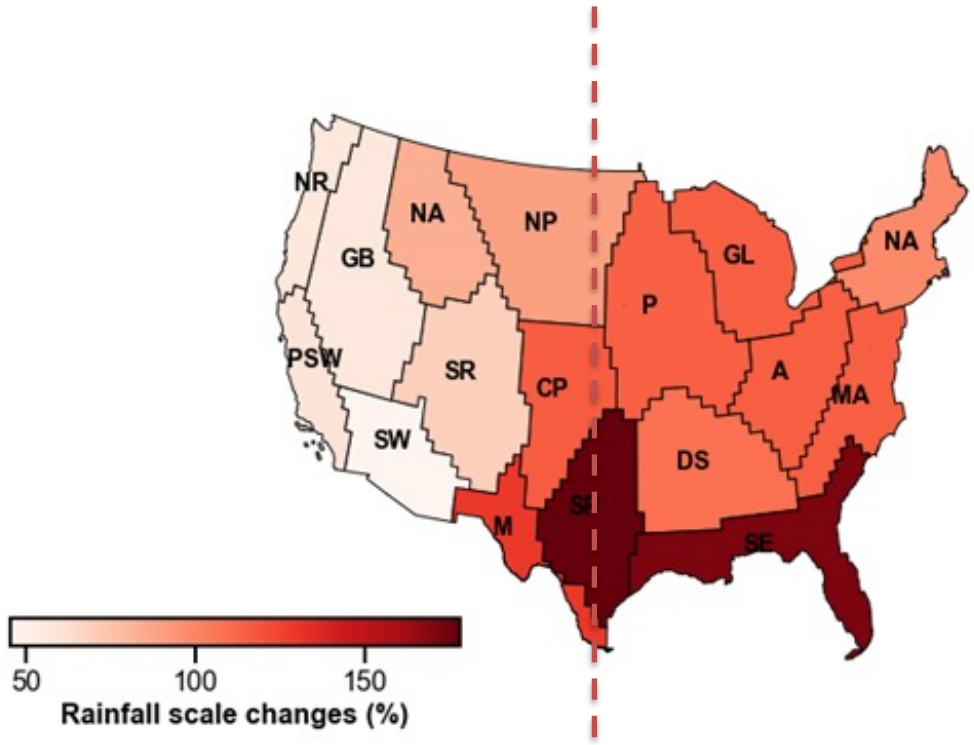
Changes in flashiness

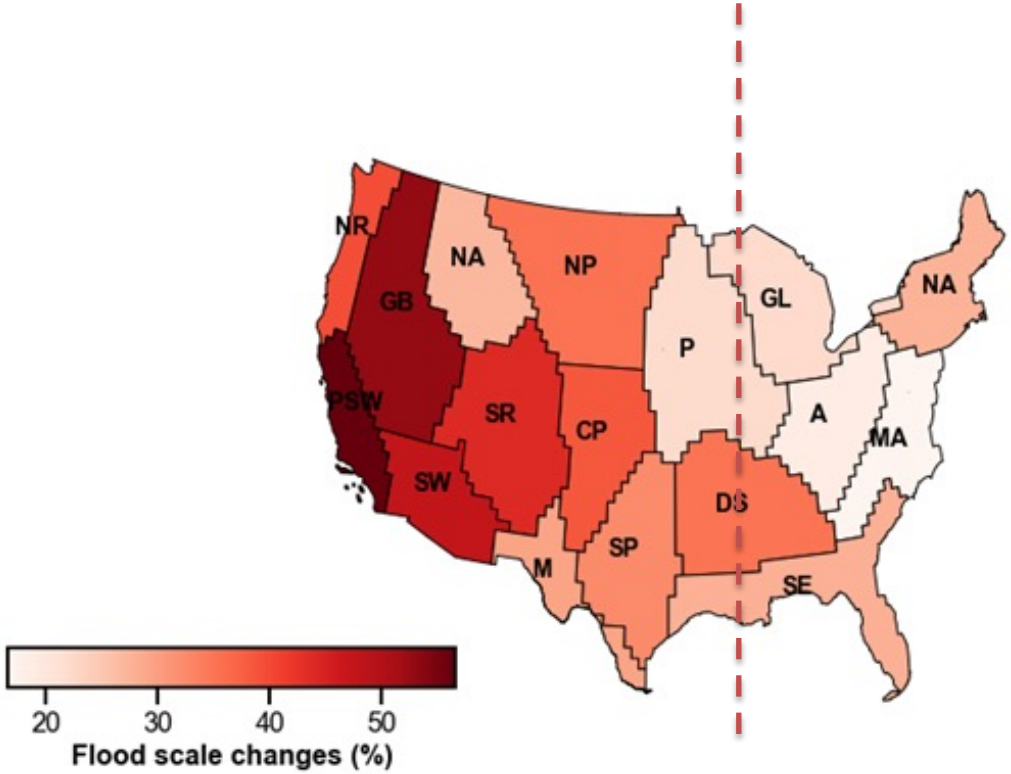


1. US floods are becoming **7.9%** flashier by the end of the century assuming a high-emissions scenario
2. The Southwest (+10.5%) has the greatest increase in flashiness among historical flash flood hot spots, and the central US (+8.6%) is emerging as a new flash flood hot spot.
3. future flash flood-prone frontiers are advancing northwards .

Distinct spatial scale changes

(b)

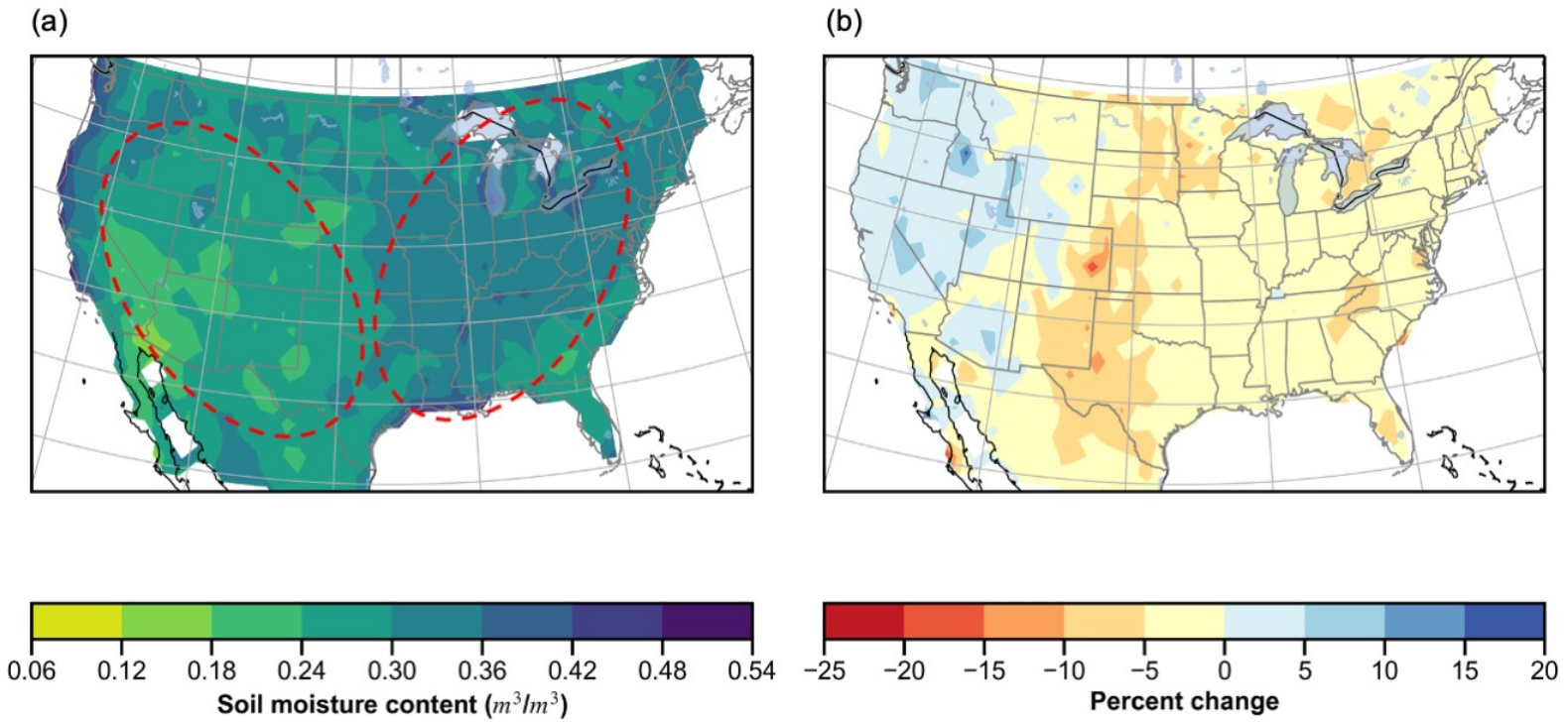




Greater changes in the East

Greater changes in the West

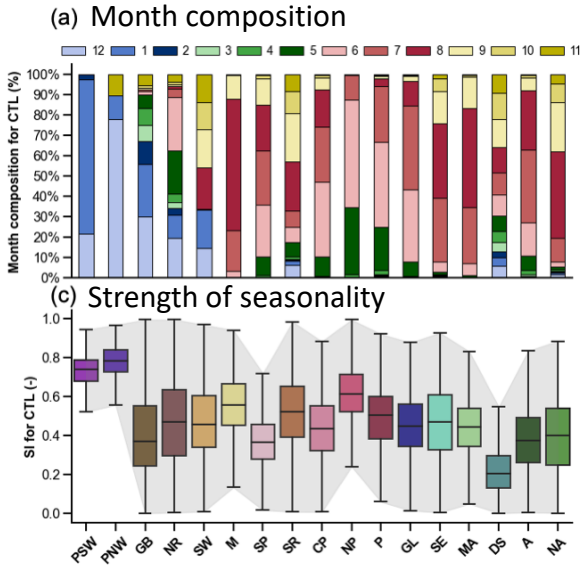
Possible reason: antecedent soil moisture

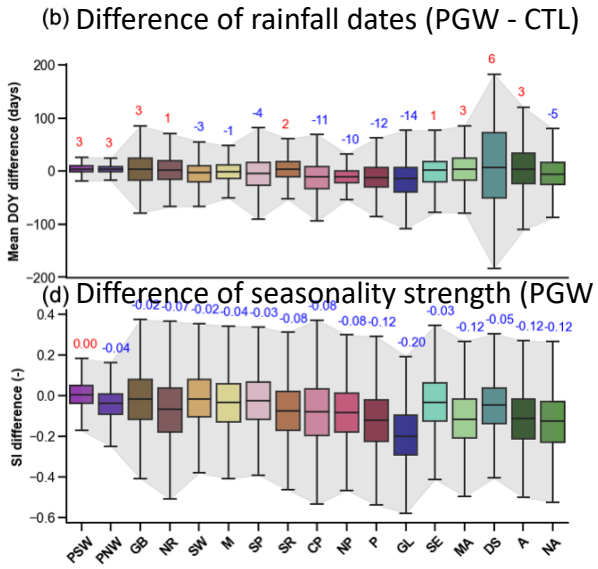


Wet antecedent soils -> accelerate flood generation
Dry antecedent soils -> slow flood generation

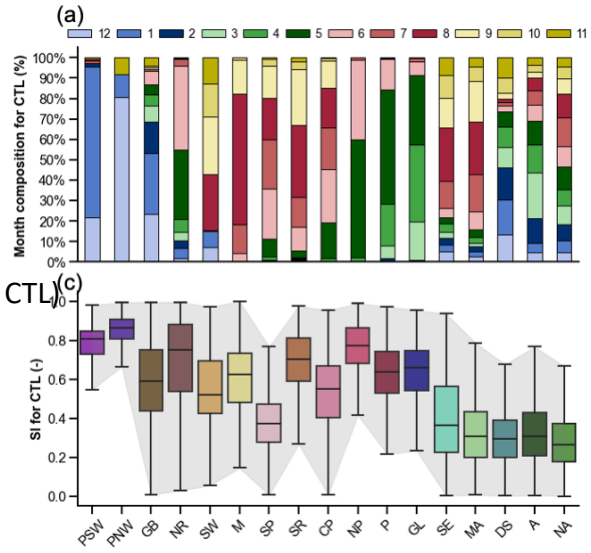
Rainfall and flood seasonality

Rainfall

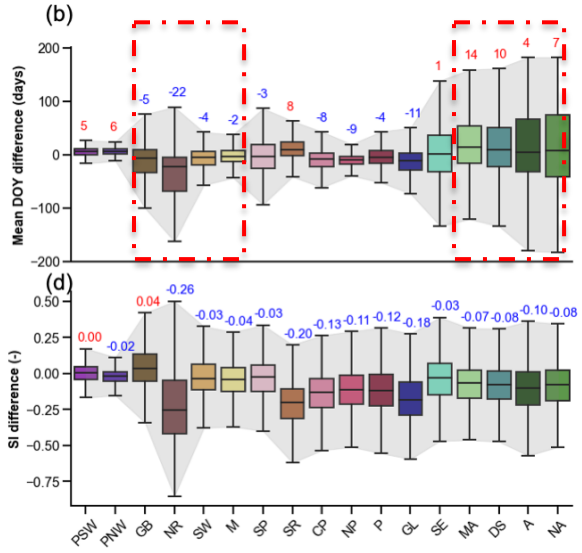




Flooding



The Rockies



West ⟶ East

Both extreme rainfall and floods in the future are becoming more random

Earlier snowmelt leads to earlier floods over the Rockies

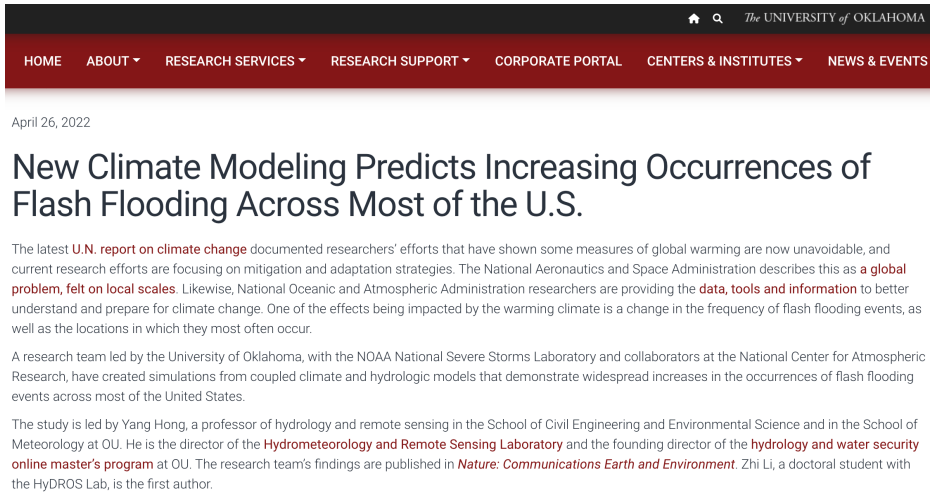
Floods in the East are delayed, due to drier antecedent soil moisture

Conclusions

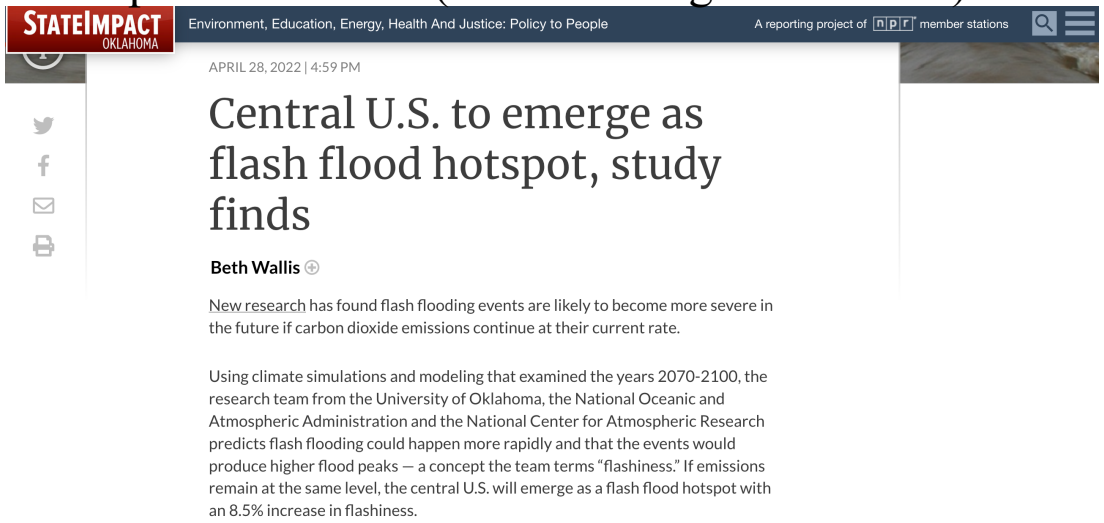
1. US floods are becoming **flashier**.
2. Extreme rainfall and floods are becoming more **frequent**, **widespread**, yet **less seasonal** (weakening seasonality)
3. Flood-generating mechanisms are changing, and we need to design our flood measures in an evolving manner to cope with those changes.

Our flood work has been featured by local, national, international news media

OUVPRP



StateImpact Oklahoma (NPR all things considered)



Independent (UK based)





The New York Times



Reference

Gao, S., Li, Zhi, Chen, M., Lin, P., Hong, Z., Allen, D., . . . Hong, Y. (2021). Spatiotemporal variability of global river extent and the natural driving factors revealed by decades of Landsat observations, GRACE gravimetry observations, and land surface model simulations. Remote Sensing of Environment, 267, 112725.

He, X., & Sheffield, J. (2020). Lagged compound occurrence of droughts and pluvials globally over the past seven decades. *Geophysical Research Letters*, 47, e2020GL087924. https://doi.org/10.1029/2020GL087924

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Li, Zhi, Gao, S., Chen, M., Gourley, J. J., & Hong, Y. (2022). Spatiotemporal characteristics of US floods: Current status and forecast under a future warmer climate. Earth's Future, 10, e2022EF002700

Li, Zhi, Chen, M., Gao, S., Luo, X., Gourley, J. J., Kirstetter, P., . . . Hong, Y. (2021). CREST-iMAP v1. 0: A fully coupled hydrologic-hydraulic modeling framework dedicated to flood inundation mapping and prediction. Environmental Modelling & Software, 141, 105051.

Li, Zhi, Tang, G., Hong, Z., Chen, M., Gao, S., Kirstetter, P., . . . Hong, Y. (2021). Two-decades of GPM IMERG early and final run products intercomparison: Similarity and difference in climatology, rates, and extremes. Journal of Hydrology, 594, 125975.

Li, Z., Gao, S., Chen, M., Gourley, J., Mizukami, N., and Hong, Y.: CREST-VEC: a framework towards more accurate and realistic flood simulation across scales, Geosci. Model Dev., 15, 6181–6196, https://doi.org/10.5194/gmd-15-6181-2022, 2022.

Maddox, R. A., Chappell, C. F., & Hoxit, L. R. (1979). Synoptic and Meso-α Scale Aspects of Flash Flood Events, *Bulletin of the American Meteorological Society*, *60*(2), 115-123.

Villarini, G., On the seasonality of flooding across the continental United States. Advances in Water Resources, 87, 80-91.

Thank you for your attention!

Questions? li1995@ou.edu